

A Certified First-Omitted-Order Ladder for the Deterministic Tail

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Abstract

We organize the direct factorwise target-tail estimate into a certified ladder of first omitted orders. Arb replay gives explicit logarithmic and multiplicative budgets for $N = 13, 21, 29, 37, 45, 53, 61$. Only $N = 29$ matches the currently archived order-28 deterministic head; higher rows are conditional interfaces, not newly constructed heads. The ladder is a target-side result and leaves the cloud bridge and uniform quotient theorem open.

1 Definition of the ladder

Let E_N denote the direct factorwise upper bound for $\sum_{n \geq N} |a_n|/n$ from RH-264 [2]. We restrict to odd N so that the even trace threshold is $(N + 1)/2$ and the odd endpoint series starts at N .

Proposition 1.1 (Monotone ladder). *For the fixed unit disk and the RH-13 certificate, the direct majorants satisfy $E_{N'} < E_N$ whenever $N' > N$ are successive listed orders. The same holds for $\exp(E_N) - 1$.*

Proof. Each component is a positive tail of an absolutely convergent majorant. The first omitted order increases, so every component decreases; the exponential map is increasing on the nonnegative reals. $\square$

2 Certified values

The 100- and 200-decimal Arb replays evaluate the same outward interval expressions independently. The resulting safe endpoints are

N	E_N	$\exp(E_N) - 1$
13	$< 2.3243607 \times 10^{-2}$	$< 2.3515845 \times 10^{-2}$
21	$< 8.73261 \times 10^{-4}$	$< 8.73642 \times 10^{-4}$
29	$< 2.6624745 \times 10^{-5}$	$< 2.6625100 \times 10^{-5}$
37	$< 9.32147 \times 10^{-7}$	$< 9.32147 \times 10^{-7}$
45	$< 4.432615 \times 10^{-8}$	$< 4.432615 \times 10^{-8}$
53	$< 1.705725 \times 10^{-9}$	$< 1.705725 \times 10^{-9}$
61	$< 7.175542 \times 10^{-11}$	$< 7.175542 \times 10^{-11}$

Theorem 2.1 (Current operational row). *The RH-253 head through order 28 and the $N = 29$ row form a certified deterministic target-side interface with logarithmic tail below 2.6624745×10^{-5} . No head through order 36 is present in the archive, so the rows $N \geq 37$ are conditional interfaces only.*

Proof. The order-28 extent is the finite dictionary of RH-253 [1]; the $N = 29$ bound is RH-264. The remaining rows are evaluated by the same analytic majorant but require coefficients not claimed here. $\square$

This ladder is not a cloud envelope, a selector, or a uniform quotient theorem. The five obligation vector remains $(0, 0, 0, 1, 1)$; all Gates A–E, Hilbert–Polya claims, zeta-divisor equality, and RH implications remain false/open.

References

- [1] Bin Wang. Extended deterministic anchor atlas. *Preprint*, 2026. RH-253 preprint.
- [2] Bin Wang. Direct factorwise deterministic-numerator tail certificate. *Preprint*, 2026. RH-264 preprint.